Biosecurity. AIs that are designed for biological research or are otherwise known to possess capabilities in biological research or engineering should be subject to increased scrutiny and access controls, since they have the potential to be repurposed for bioterrorism. In addition, system developers should research and implement methods to remove biological data from the training dataset or excise biological capabilities from finished systems, if those systems are intended for general use [14]. Researchers should also investigate ways that AIs could be used for biodefense, for example by improving biological monitoring systems, keeping in mind the potential for dual use of those applications. In addition to AI-specific interventions, more general biosecurity interventions can also help mitigate risks. These interventions include early detection of pathogens through methods like wastewater monitoring [28], far-range UV technology, and improved personal protective equipment [6].

Restricted access. AIs might have dangerous capabilities that could do significant damage if used by malicious actors. One way to mitigate this risk is through structured access, where AI providers limit users' access to dangerous system capabilities by only allowing controlled interactions with those systems through cloud services [29] and conducting know your customer screenings before providing access [30]. Other mechanisms that could restrict access to the most dangerous systems include the use of hardware, firmware, or export controls to restrict or limit access to computational resources [31]. Lastly, AI developers should be required to show that their AIs pose minimal risk of catastrophic harm prior to open sourcing them. This recommendation should not be construed as permitting developers to withhold useful and non-dangerous information from the public, such as transparency around training data necessary to address issues of algorithmic bias or copyright.

Technical research on adversarially robust anomaly detection. While preventing the misuse of AIs is critical, it is necessary to establish multiple lines of defense by detecting misuse when it does happen. AIs could enable anomaly detection techniques that could be used for the detection of unusual behavior in systems or internet platforms, for instance by detecting novel AI-enabled disinformation campaigns before they can be successful. These techniques need to be adversarially robust, as attackers will aim to circumvent them.

Legal liability for developers of general-purpose AIs. General-purpose AIs can be fine-tuned and prompted for a wide variety of downstream tasks, some of which may be harmful and cause substantial damage. AIs may also fail to act as their users intend. In either case, developers and providers of general-purpose systems may be best placed to reduce risks, since they have a higher level of control over the systems and are often in a better position to implement mitigations. To provide strong incentives for them to do this, companies should bear legal liability for the actions of their AIs. For example, a strict liability regime would incentivize companies to minimize risks and purchase insurance, which would cause the cost of their services to more closely reflect externalities [32]. Regardless of what liability regime is ultimately used for AI, it should be designed to hold AI companies liable for harms that they could have averted through more careful development, testing, or standards [33].

Positive Vision

In an ideal scenario, it would be impossible for any individual or small group to use AIs to cause catastrophes. Systems with extremely dangerous capabilities either would not exist at all or would be controlled by a democratically accountable body committed to using them only for the general wellbeing of the population. Like nuclear weapons, the information needed to develop those capabilities would remain carefully guarded to prevent proliferation. At the same time, control of AI systems would be subject to strong checks and balances, avoiding entrenchment of power inequalities. Monitoring tools would be utilized at the minimum level necessary to make risks negligible and could not be used to suppress dissent.

3 AI Race

The immense potential of AIs has created competitive pressures among global players contending for power and influence. This "AI race" is driven by nations and corporations who feel they must rapidly build and deploy AIs to secure their positions and survive. By failing to properly prioritize global risks, this dynamic makes it more likely that AI development will produce dangerous outcomes. Analogous to the nuclear arms race during the Cold War, participation in an AI race may serve individual short-term interests, but it ultimately results in worse collective outcomes for humanity. Importantly, these risks stem not only from the intrinsic nature of AI technology, but from the competitive pressures that encourage insidious choices in AI development.

In this section, we first explore the military AI arms race and the corporate AI race, where nation-states and corporations are forced to rapidly develop and adopt AI systems to remain competitive. Moving beyond these specific races, we reconceptualize competitive pressures as part of a broader evolutionary process in which AIs could become increasingly pervasive, powerful, and entrenched in society. Finally, we highlight potential strategies and policy suggestions to mitigate the risks created by an AI race and ensure the safe development of AIs.

3.1 Military AI Arms Race

The development of AIs for military applications is swiftly paving the way for a new era in military technology, with potential consequences rivaling those of gunpowder and nuclear arms in what has been described as the "third revolution in warfare." The weaponization of AI presents numerous challenges, such as the potential for more destructive wars, the possibility of accidental usage or loss of control, and the prospect of malicious actors co-opting these technologies for their own purposes. As AIs gain influence over traditional military weaponry and increasingly take on command and control functions, humanity faces a paradigm shift in warfare. In this context, we will discuss the latent risks and implications of this AI arms race on global security, the potential for intensified conflicts, and the dire outcomes that could come as a result, including the possibility of conflicts escalating to a scale that poses an existential threat.

3.1.1 Lethal Autonomous Weapons (LAWs)

LAWs are weapons that can identify, target, and kill without human intervention [34]. They offer potential improvements in decision-making speed and precision. Warfare, however, is a high-stakes, safety-critical domain for AIs with significant moral and practical concerns. Though their existence is not necessarily a catastrophe in itself, LAWs may serve as an on-ramp to catastrophes stemming from malicious use, accidents, loss of control, or an increased likelihood of war.

Figure 7: Low-cost automated weapons, such as drone swarms outfitted with explosives, could autonomously hunt human targets with high precision, performing lethal operations for both militaries and terrorist groups and lowering the barriers to large-scale violence.

LAWs may become vastly superior to humans. Driven by rapid developments in AIs, weapons systems that can identify, target, and decide to kill human beings on their own—without an officer directing an attack or a soldier pulling the trigger—are starting to transform the future of conflict. In 2020, an advanced AI agent outperformed experienced F-16 pilots in a series of virtual dogfights, including decisively defeating a human pilot 5-0, showcasing "aggressive and precise maneuvers the human pilot couldn't outmatch" [35]. Just as in the past, superior weapons would allow for more destruction in a shorter period of time, increasing the severity of war.

Militaries are taking steps toward delegating life-or-death decisions to AIs. Fully autonomous drones were likely first used on the battlefield in Libya in March 2020, when retreating forces were "hunted down and remotely engaged" by a drone operating without human oversight [36]. In May 2021, the Israel Defense Forces used the world's first AI-guided weaponized drone swarm during combat operations, which marks a significant milestone in the integration of AI and drone technology in warfare [37]. Although walking, shooting robots have yet to replace soldiers on the battlefield, technologies are converging in ways that may make this possible in the near future.

LAWs increase the likelihood of war. Sending troops into battle is a grave decision that leaders do not make lightly. But autonomous weapons would allow an aggressive nation to launch attacks without endangering the lives of its own soldiers and thus face less domestic scrutiny. While remote-controlled weapons share this advantage, their scalability is limited by the requirement for human operators and vulnerability to jamming countermeasures, limitations that LAWs could overcome [38]. Public opinion for continuing wars tends to wane as conflicts drag on and casualties increase [39]. LAWs would change this equation. National leaders would no longer face the prospect of body bags returning home, thus removing a primary barrier to engaging in warfare, which could ultimately increase the likelihood of conflicts.

3.1.2 Cyberwarfare

As well as being used to enable deadlier weapons, AIs could lower the barrier to entry for cyberattacks, making them more numerous and destructive. They could cause serious harm not only in the digital environment but also in physical systems, potentially taking out critical infrastructure that societies depend on. While AIs could also be used to improve cyberdefense, it is unclear whether they will be most effective as an offensive or defensive technology [40]. If they enhance attacks more than they support defense, then cyberattacks could become more common, creating significant geopolitical turbulence and paving another route to large-scale conflict.

AIs have the potential to increase the accessibility, success rate, scale, speed, stealth, and potency of cyberattacks. Cyberattacks are already a reality, but AIs could be used to increase their frequency and destructiveness in multiple ways. Machine learning tools could be used to find more critical vulnerabilities in target systems and improve the success rate of attacks. They could also be used to increase the scale of attacks by running millions of systems in parallel, and increase the speed by finding novel routes to infiltrating a system. Cyberattacks could also become more potent if used to hijack AI weapons.

Cyberattacks can destroy critical infrastructure. By hacking computer systems that control physical processes, cyberattacks could cause extensive infrastructure damage. For example, they could cause system components to overheat or valves to lock, leading to a buildup of pressure culminating in an explosion. Through interferences like this, cyberattacks have the potential to destroy critical infrastructure, such as electric grids and water supply systems. This was demonstrated in 2015, when a cyberwarfare unit of the Russian military hacked into the Ukrainian power grid, leaving over 200,000 people without power access for several hours. AI-enhanced attacks could be even more devastating and potentially deadly for the billions of people who rely on critical infrastructure for survival.

Difficulties in attributing AI-driven cyberattacks could increase the risk of war. A cyberattack resulting in physical damage to critical infrastructure would require a high degree of skill and effort to execute, perhaps only within the capability of nation-states. Such attacks are rare as they constitute an act of war, and thus elicit a full military response. Yet AIs could enable attackers to hide their identity, for example if they are used to evade detection systems or more effectively cover the tracks of the attacker [41]. If cyberattacks become more stealthy, this would reduce the threat of retaliation from an attacked party, potentially making attacks more likely. If stealthy attacks do happen, they might incite actors to mistakenly retaliate against unrelated third parties they suspect to be responsible. This could increase the scope of the conflict dramatically.